
SESSION REPORT

WEEK-1

CRISPR Basics

- The origin of CRISPR as a defense mechanism in bacteria
- The CRISPR-Cas9 mechanism for gene editing
- What makes CRISPR significant? Its effects on genome modification

DNA Repair and Editing Results

- Pathways for repairing DNA double-strand breaks
- Different types of editing results
- Factors that affect editing efficiency

CRISPR's Role in Current Research

- Uses in contemporary biology
- Applications in the medical field
- Ethical and regulatory issues to consider

WEEK-2

Gene Structures & Coordinates

- Comprehending genome organization
- Genome coordinate frameworks
- Utilizing genome browsers

Extracting Target Sequences

- Identifying editing regions
- Obtaining DNA sequences
- Preparing sequences for CRISPR design

Designing Editing Strategy

- Establishing the experiment objective
- Choosing the editing site
- Planning subsequent validation

WEEK-3

PAM & gRNA Rules

- Requirements for PAM
- Structure of guide RNA
- Sequence guidelines for effective guides

Guide Design Tools

- Overview of automated design tools
- Interpreting tool outputs
- Exporting guide options

Comparative Guide Selection

- Evaluating candidate guides
- Balancing specificity and efficiency
- Choosing final guides

WEEK-4

Off-Target Mechanisms

- The occurrence of off-target binding
- Biological implications
- Factors that affect off-target occurrences

Off-Target Prediction Tools

- Genome-wide analysis for similar sequences
- Assessing off-target impact
- Identifying risky guides

Final Guide Optimization

- Combining all design criteria
- Selecting final experimental guides
- Preparing sequences for synthesis or cloning

WEEK-5

Introduction to CRISPR Sequencing Data

- The necessity of sequencing
- Different types of sequencing data
- Comprehending FASTQ files

QC and Alignment

- Quality control and sequencing
- Alignment of reads
- Understanding alignment results

Indel Detection & Editing Efficiency

- Identifying insertions and deletions
- Measuring editing effectiveness
- Analysis of mutation patterns

WEEK 6

- Understanding the biological impact of CRISPR alterations and evaluating their outcomes.
- Examining editing effectiveness and indel profiles in a biological setting
- Using simple plots and summaries to visualize CRISPR results

Advanced Reporting and Visualization

- Developing sophisticated CRISPR data visualizations and comparison analysis
- Organizing findings into an understandable and methodical CRISPR analysis process
- Writing a comprehensive report that includes final project observations and interpretation